

Project Proposal

A Comparative Life Cycle Cost Analysis of Sustainable and Conventional Buildings

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IMPROVED LITERATURE REVIEW:

Life Cycle Costing and Sustainable Construction

An economic assessment technique called life cycle costing (LCC) is used to determine how much a structure will cost overall over the lifetime, including the expenses associated with initial construction, operation, maintenance, replacement, and disposal. According to research, operational costs make up between 60 and 80 percent of all building life cycle costs, making operational effectiveness a crucial component of long term economic performance.

Sustainable buildings are made to use less energy and water while reducing their negative effects on the environment. Several studies have shown significant long-term operational savings, despite the fact that sustainable buildings typically require 5–15% greater initial construction costs than conventional buildings. Green buildings can cut energy use by about 25–30%, according to research by Zuo and Zhao (2017), while sustainable buildings can cut greenhouse gas emissions by up to 50%, according to GBCA data.

According to research by Cabeza et al. (2014), using LCC techniques increases decision-making accuracy in building projects by over 15% to 20%.

Additionally, sustainable structures accomplish:

- 20 – 40% reduction in energy usage
- 30 – 50% reduction in water use
- 10 – 25% reduction in maintenance expenses

Over the course of a building's 30–50 year lifespan, these reductions can lead to a 10–20% reduction in total life cycle expenses.

Cost Optimization and Industry Importance

Long-term cost planning is becoming more crucial since the Australian construction sector has seen yearly cost rises of between 6–10%. Sustainable design strategies assist environmental goals while lowering operating costs. Over 75% of newly built homes in New South Wales now meet 7-star energy efficiency criteria, indicating a growing trend in the use of sustainable building techniques.

Important financial factors influencing LCC results include:

- Rates of discount (4 – 7%)
- Increase in energy prices (2 – 5% each year)
- Lifespan of construction (30 – 50 years)
- Material durability and maintenance cycles

Long-term financial results and overall cost computations are greatly impacted by the sensitivity of these variables.

Research Gap:

There is limited study of Sydney specific data in existing research, which mostly concentrate on worldwide or generic Australian data. Few studies combine sustainable performance metrics with realistic Quantity Surveying (QS) cost data. Furthermore, there is ambiguity when comparing the performance of conventional and sustainable buildings due to varying assumptions about energy escalation and discount rates.

By performing a comparative LCC assessment with a particular focus on Sydney construction conditions and using consistent financial assumptions for analysis, this study seeks to overcome these constraints.

Primary Research Question:

In terms of life cycle cost performance in Sydney construction and operating conditions, how do sustainable and conventional buildings compare?

Sub Questions:

- a) In comparison to conventional structures, how much do sustainable buildings save operating costs?
- b) What effects do rising energy prices and discount rates have on life cycle costing results?
- c) How much does the overall life cycle cost of conventional and sustainable buildings differ by percentage?
- d) In sustainable buildings, which cost elements have the biggest impact on long-term savings?
- e) How might life cycle costing aid in the planning of sustainable infrastructure and construction?

RESEARCH METHODOLOGY PROPOSAL:

Using sensitivity analysis, life cycle costing computations, and secondary data analysis, this study employs a comparative analytical methodology.

Phase 1: Data Collection and Assumption Setting

Gather secondary data from peer-reviewed publications, Infrastructure Australia market capacity studies, Green Building Council of Australia (GBCA) papers, and building price insights from the Australian Bureau of Statistics (ABS). Set baseline presumptions:

Paramater	Assumption
Building lifespan	40 years
Discount rate	5% (sensitivity: 4–7%)
Energy price escalation	3% annually (sensitivity: 2–5%)
Sustainable building premium	+5–15% initial cost
Energy savings	20–40%
Water savings	30–50%
Maintenance savings	10–25%

Justification: Without the time and expense of primary construction projects, secondary data enables comparison of current real-world building performance. Assumptions are derived from industry norms and verified literature.

Phase 2: Life Cycle Cost Calculation

Calculate total LCC for sustainable and conventional building models using the formula:

$$Lcc = \text{Initial Cost} + \sum \text{Annual Operational Cost} \times \text{Discount Factor} + \text{Maintenance Cost} + \text{Disposal Cost}$$

Compare cost components:

Cost Component	Sustainable Building	Conventional Building
Initial Cost	+ 10%	Base value
Energy Cost	25 – 40% lower	Higher
Water Cost	30 – 50% lower	Higher
Maintenance	10 – 20% lower	Moderate
Total LCC (40 yrs)	10 – 20% lower	Higher

The findings of a structured LCC computation are quantitative, comparable, and sensitive to financial assumptions.

Phase 3: Sensitivity Analysis

Examine how changes in important factors impact overall LCC results:

Parameter	Range Tested	Baseline
Discount rate	4%, 5%, 6%, 7%	5%
Energy price escalation	2%, 3%, 4%, 5%	3%
Building lifespan	30, 40, 50 years	40 years

Sensitivity analysis improves transparency and enables decision-makers to apply findings under many economic situations by identifying which assumptions have the greatest impact on outcomes.

Phase 4: Comparative Analysis and Recommendations

- Calculate the percentage variations in total LCC between different types of buildings.
- Determine the break-even points at which higher initial costs are covered by operational savings.
- Make suggestions for Sydney's industry and policy stakeholders.
- Describe the limitations and suggest future lines of inquiry.

Recommendations that are useful connect academic research with practical applications in the government, real estate, and construction industries.

Ethical and Indigenous Dimensions of Research:

Ethical Considerations:

- Every secondary data source shall be appropriately cited and referenced.
- Throughout the entire study process, academic integrity shall be upheld.
- Calculations and financial data will be openly recorded.
- There will be no usage of private or human participant data.

Environmental Sustainability:

By calculating the financial advantages of green buildings, this study advances sustainability objectives. Buildings consume over 40% of the world's energy; proving cost-effectiveness can hasten the adoption of sustainable design and help reduce greenhouse gas emissions by 30–50% (GBCA, 2023).

Indigenous Knowledge and Sustainability:

Based on in-depth ecological knowledge, Indigenous Australians have long engaged in sustainable land and resource management. Indigenous environmental stewardship values are respected in this study by:

- Realizing that Indigenous principles of intergenerational care for land and resources are consistent with sustainable building
- Recognizing that water and energy efficiency lessens the environmental impact on traditional lands
- Noting that Indigenous communities that are disproportionately impacted by climate change benefit from government sustainability goals (such as emissions reductions).

No Indigenous cultural knowledge or materials are directly used in this secondary-data study.

Safety and Risk:

There is no actual laboratory work involved. There is little risk to safety because all work is done at a desk. We'll adhere to standard academic procedures for cyber security and data management.

Research Project Plan, Milestones, and Timeline:

Phase	Time Period	Activities	Milestones	Deliverables
Phase 1	Weeks 1–3	Literature review and data collection	Completion of literature review	Literature summary and data sources
Phase 2	Weeks 4–6	Development of LCC models	Initial LCC calculations completed	Comparative cost tables
Phase 3	Weeks 7–9	Sensitivity analysis	Validation of financial assumptions	Sensitivity analysis graphs
Phase 4	Weeks 10–12	Comparative evaluation and report writin	Final analysis completed	Final research report and recommendations

Risk Management:

Risk	Likelihood	Impact	Mitigation Strategy
Limited Sydney specific cost data	Medium	High	Use multiple Australian government and industry sources
Inconsistent literature findings	Medium	Medium	Apply standardized assumptions for comparison
Errors in financial calculation	Low	High	Cross-check calculations using multiple methods
Time management delays	Medium	Medium	Use milestone-based project scheduling
Uncertainty in future energy prices	High	Medium	Conduct sensitivity analysis

Anticipated Peer Review Concerns and Responses:

Concern 1: Minimal utilization of primary data:

To ensure dependability and a comprehensive comparative study, the research relies on peer-reviewed papers and confirmed secondary industry data.

Concern 2: Sensitivity of LCC presumptions:

To increase robustness, sensitivity analysis will be carried out utilizing different discount rates and energy escalation scenarios.

Concern 3: Limited information unique to Sydney

To increase local relevance, NSW construction data and Australian government reports will be included.

Concern 4: Inability to forecast long term expenses

The study uses scenario-based analysis to overcome the limitation of long-term forecasting uncertainty.

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Appendix A: Communication Log with Research Adviser

Date	Meeting Format	Discussion Points	Action Items	Follow Up Date
April 10, 2026	Email	Discussed project topic and scope of Life Cycle Costing analysis	Begin literature review and collect sustainability studies	May 5, 2026
May 5, 2026	Online Meeting	- Reviewed literature review draft and identified research gaps - Discussed LCC assumptions and operational cost factors	- Include Sydney-specific construction cost data - Develop comparative cost framework	May 18, 2026
May 18, 2026	Email	Discussed final proposal structure and recommendations	Complete formatting and referencing	May 18, 2026